Letter

Efficient spin Seebeck and spin Nernst effects of magnons in altermagnets

Qirui Cui,[1,3,4] Bowen Zeng,[2] Ping Cui,[3,4] Tao Yu,[2,*] and Hongxin Yang[1,†]

[1]*National Laboratory of Solid State Microstructures, School of Physics, Collaborative Innovation Center of Advanced Microstructures, Nanjing University, Nanjing 210093, China*

[2]*School of Physics, Huazhong University of Science and Technology, Wuhan 430074, China*

[3]*Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences, Ningbo 315201, China*

[4]*Yongjiang Laboratory, Ningbo 315202, China*

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We report two nondegenerate magnon modes with opposite spins or chiralities in collinearly antiferromagnetic insulators driven by symmetry-governed anisotropic exchange couplings. The consequent giant spin splitting contributes to spin Seebeck and spin Nernst effects generating longitudinal and transverse spin currents when the temperature gradient applies along and away from the main crystal axis, without requiring any external magnetic field and spin-orbit coupling. Based on first-principle calculations, we predict feasible material candidates holding robust altermagnetic spin configurations to efficiently transport spin. The spin Seebeck conductivity is comparable to the records of antiferromagnets that require the magnetic field, and the spin Nernst conductivity is two orders in magnitude larger than that in antiferromagnetic monolayers that need Berry curvature.

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Introduction. Magnons in magnetic insulators favor the spin transport with low energy consumption [1–11]. Although the chirality of magnon is fixed in ferromagnets, symmetry allows two degenerate magnon modes carrying opposite spins in collinear antiferromagnets with easy-axis anisotropy [12–17]. With no net magnetization, vanished static stray field, and terahertz response rate, antiferromagnets exhibit unique advantages for robust and ultrafast spin-based nanoscale applications, among which the spin Seebeck effect (SSE) and spin Nernst effect (SNE) are, respectively, effective approaches to longitudinally and transversely transport spins with respect to the temperature gradient [13,15,17–35]. In collinear antiferromagnet, the SSE often requires a magnetic field to break the spin degeneracy or enhance the spin splitting [12,13,26,36], and more stringently, the SNE exploits the Berry curvature that needs the Dzyaloshinskii-Moriya interaction (DMI) [15,27,37,38] or magnetoelastic coupling [33,34], i.e., the transverse spin current is normally proportional to the spin-orbit coupling (SOC) strength. Theoretical effort made to realize spin transport free of external magnetic field and SOC is based on the noncollinear antiferromagnetic insulator of kagome lattice [39]. Nevertheless, to ensure the identical spin configurations between kagome layers, a large magnetic field appears to be necessary. Also, the symmetry in such kagome lattice still allows for the nearest-neighboring DMI, which is used to modulate the spin transport [40]. Therefore it is fair to say the spin transport via magnons, which can be *entirely* free of the magnetic field and Berry curvature, in antiferromagnets remains wanting.

Metallic altermagnet is a special type of collinear antiferromagnet without net magnetization that was recently reported to hold electronic bands with spin splitting in the reciprocal space, even in the absence of any relativistic SOC, due to breaking of the combined space inversion, time reversal, and translation symmetries [41–47]. By applying the external electrical field, the motion of the Fermi surface with spin splitting could result in the strong transverse spin current, similar to the spin Hall effect, which was experimentally demonstrated based on the altermagnetic metal RuO_2 [48–50]. Interestingly, antiferromagnetic magnons possessing the nondegenerate chirality have recently been predicted in RuO_2 as well [51]. However, the spin transport properties dominated by such magnons are still unclear, for which the insulators can avoid the complications from electrons in metals.

In this Letter, we report two spin nondegenerate α and β magnon modes allowed by symmetry in collinear antiferromagnets, driven by the anisotropic spin exchange couplings [Fig. 1(a)]. This leads to the anisotropic spin-momentum locking for magnons favoring efficient SSE and SNE, which are free of external magnetic fields or Berry curvature. Specifically, nonequivalent magnon currents carried by α and β modes with opposite spins are stimulated by the temperature gradient but flow along different directions, which makes their combination produce longitudinal and transverse spin currents [Fig. 1(b)]. Based on first-principle calculations, we propose potential material candidates, e.g., $Cr_2Te(Se)_2O$ monolayer, for observing these phenomena. Our findings offer the missing piece of spin transport in altermagnets by opening a novel way towards producing spin current via antiferromagnetic magnons and providing an interesting platform for exploring spin caloritronic effects.

Model. We construct a two-dimensional antiferromagnetic square lattice with nonrelativistic spin splitting by symmetry

*taoyuphy@hust.edu.cn
†hongxin.yang@nju.edu.cn

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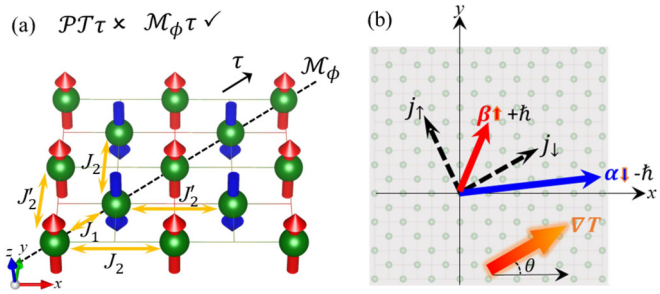


FIG. 1. (a) The schematic of two-dimensional antiferromagnetic spin lattice, without $\mathcal{PT}\tau$ symmetry but holding $\mathcal{M}_\phi\tau$ symmetry. The red and blue arrows represent spin vectors in the out-of-plane orientation. (b) A thermal gradient away from the main axis drives longitudinal and transverse spin currents, contributed by different magnon modes with opposite spins.

analysis [Fig. 1(a)]. The lacking of combined space inversion $\mathcal{P}$, time reversal $\mathcal{T}$ and half-unit cell translation τ symmetries ($\mathcal{PT}\tau$) induces the spin-splitting for electronic eigenstates, while the preservation of combined mirror $\mathcal{M}_\phi$ and τ symmetries ($\mathcal{M}_\phi\tau$) guarantees eigenstates holding opposite spins at $\mathbf{k}$ and $\mathcal{M}_\phi\mathbf{k}$, thus tuning the conventional antiferromagnet to altermagnet [44,52,53]. Notably, from the structural aspect, $\mathcal{PT}\tau$ symmetry breaking requires the nearest-neighboring spin sites in each sublattice being connected anisotropically in the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ directions. Thereby, the bridged electronic states give rise to anisotropic superexchange coupling based on Goodenough-Kanamori-Anderson (GKA) theory [54–56]. Further, $\mathcal{M}_\phi\tau$ symmetry requires spin-connected types of sublattices "a" and "b" being exactly swapped in $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, and Néel magnetization paralleling with $\mathcal{M}_\phi$, which results in the switched exchange coupling for two sublattices. The symmetry-governed spin Hamiltonian

$$\begin{aligned}\hat{H} =& J_1 \sum_{\langle i,j\rangle} \hat{\mathbf{S}}_{ai}\cdot\hat{\mathbf{S}}_{bj} \\ &+ J_2 \sum_{\langle i_x,j_x\rangle} \hat{\mathbf{S}}_{ai}\cdot\hat{\mathbf{S}}_{aj} + J_2' \sum_{\langle i_y,j_y\rangle} \hat{\mathbf{S}}_{ai}\cdot\hat{\mathbf{S}}_{aj} + K\sum_i \left(\hat{S}_{ai}^z\right)^2 \\ &+ J_2' \sum_{\langle i_x,j_x\rangle} \hat{\mathbf{S}}_{bi}\cdot\hat{\mathbf{S}}_{bj} + J_2 \sum_{\langle i_y,j_y\rangle} \hat{\mathbf{S}}_{bi}\cdot\hat{\mathbf{S}}_{bj} + K\sum_i \left(\hat{S}_{bi}^z\right)^2\end{aligned} \tag{1}$$

describes explicitly these magnetic properties, where $\hat{\mathbf{S}}_a$ and $\hat{\mathbf{S}}_b$ represent the spin operators of the same magnitude. $J_1 > 0$ favors the antiferromagnetic inter-sublattice exchange coupling, $J_2(J_2') < 0$ favors the ferromagnetic intrasublattice exchange coupling, and single-ion magnetic anisotropy $K < 0$ favors the out-of-plane magnetization. $\langle i, j\rangle$, $\langle i_x, j_x\rangle$, and $\langle i_y, j_y\rangle$ indicates the nearest-neighboring coupling along the diagonal, $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, respectively.

Using the Holstein-Primakoff transformation [57], the Hamiltonian in terms of bosonic operators $\hat{\psi}_\mathbf{k} = (\hat{a}_\mathbf{k}, \hat{b}_\mathbf{k}^\dagger)^T$ in the momentum space reads $\hat{H} = \sum_\mathbf{k} \hat{\psi}_\mathbf{k}^\dagger \hat{H}_\mathbf{k} \hat{\psi}_\mathbf{k}$, where

$$\hat{H}_\mathbf{k} = S\begin{pmatrix} J^* + 2J_2\gamma_1 + 2J_2'\gamma_2 & 4J_1\gamma_3 \\ 4J_1\gamma_3 & J^* + 2J_2'\gamma_1 + 2J_2\gamma_2 \end{pmatrix}. \tag{2}$$

Here the on-site energy $J^* = 4J_1 - 2J_2 - 2J_2' - 2K$, and the structure factors $\gamma_1 = \cos(k_x a)$, $\gamma_2 = \cos(k_y a)$, $\gamma_3 = \cos(k_x a/2)\cos(k_y a/2)$. This Hamiltonian is diagonalized by the Bogoliubov transformation $\hat{\psi}_\mathbf{k} = T_\mathbf{k}\hat{\Psi}_\mathbf{k}$ in terms of the magnon operators of α and β modes $\hat{\Psi}_\mathbf{k} = (\hat{\alpha}_\mathbf{k}, \hat{\beta}_\mathbf{k}^\dagger)^T$:

$$\hat{H} = \sum_\mathbf{k} \hat{\Psi}_\mathbf{k}^\dagger \begin{pmatrix} \hbar\omega_\mathbf{k}^\alpha & 0 \\ 0 & \hbar\omega_\mathbf{k}^\beta \end{pmatrix} \hat{\Psi}_\mathbf{k}, \tag{3}$$

where the eigenfrequencies of α and β modes

$$\begin{aligned}\omega_\mathbf{k}^\alpha &= \frac{S}{\hbar}\left[(J_2 - J_2')(\gamma_1 - \gamma_2) + \frac{4J_1\gamma_3\Delta}{\gamma_e}\right], \\ \omega_\mathbf{k}^\beta &= \frac{S}{\hbar}\left[(J_2' - J_2)(\gamma_1 - \gamma_2) + \frac{4J_1\gamma_3\Delta}{\gamma_e}\right].\end{aligned} \tag{4}$$

Here $\Delta = \sqrt{1-{\gamma_e}^2}$ and effective form factor $\gamma_e = 4J_1\gamma_3/[4J_1 - 2K - (J_2 + J_2')(2 - \gamma_1 - \gamma_2)]$. Importantly, α and β modes are nondegenerate in the reciprocal space, except for $\mathrm{M} \leftrightarrow \Gamma$ lines at which $(\gamma_1 - \gamma_2) = 0$. The corresponding frequency splitting

$$\Delta\omega = \frac{2S}{\hbar}|(J_2 - J_2')(\gamma_1 - \gamma_2)|. \tag{5}$$

$J_2 \neq J_2'$ in each sublattice is induced by the $\mathcal{PT}\tau$ symmetry breaking, while J_2-J_2' swapping is protected by the $\mathcal{M}_\phi\tau$ symmetry, which render such mode splitting. Besides spectrum splitting, α and β modes exhibit opposite chiralities and spins. Via Bogoliubov transformation, the creation operators of α and β modes $\hat{\alpha}_\mathbf{k}^\dagger = u_\mathbf{k}\hat{a}_\mathbf{k}^\dagger - v_\mathbf{k}\hat{b}_\mathbf{k}$ and $\hat{\beta}_\mathbf{k}^\dagger = u_\mathbf{k}\hat{b}_\mathbf{k}^\dagger - v_\mathbf{k}\hat{a}_\mathbf{k}$. $\hat{S}_i^+$ and $\hat{S}_i^-$ generate opposite spin precessions at site i, which are switched between sublattices "a" and "b." Accordingly, the dynamics of $\hat{\alpha}_\mathbf{k}^\dagger$ ($\hat{\beta}_\mathbf{k}^\dagger$) is governed by right-handed (left-handed) circular polarization in "a" and "b." For α and β magnon in antiferromagnets with out-of-plane magnetization, the $\hat{\mathbf{z}}$-component spin angular momentum with unit of $\hbar$ is opposite since $\langle 0|\hat{\alpha}_k S^z \hat{\alpha}_k^\dagger|0\rangle = -1$ and $\langle 0|\hat{\beta}_k S^z \hat{\beta}_k^\dagger|0\rangle = 1$. Our model thus elucidates the basic properties of altermagnetic magnons, including spectrum splitting, chirality, and chirality-locked spin.

The out-of-plane magnetization also opens up a magnon excitation gap of $(2S/\hbar)\sqrt{K^2 - 4J_1K}$ at Γ point, which is crucial for stabilizing two-dimensional magnetism with long-range order by overcoming restrictions of Mermin-Wanger theorem [58–60]. We detail the construction of magnon Hamiltonian in Ref. [61].

Material candidates. Next, we show chromium oxytelluride, Cr_2Te_2O (CTO) monolayer [see crystal structure in Fig. 2(a)] as a promising insulator for transporting altermagnetic magnons, referring to SM [61] for details of the first-principle calculations. CTO possesses the $P4/mmm$ space group, and the existence of bridging Te and O between Cr breaks $\mathcal{PT}\tau$ symmetry [Fig. 2(a)]. The octahedral crystal filed around Cr approximately splits $3d$ orbitals into the lower energy set t_{2g} and higher energy set e_g. Accordingly, each Cr carries a net magnetic moment around $3.29\mu_B$, which is consistent with the spin number $S = 3/2$ of Cr^{3+}. By using energy mapping methods, we calculate $J_1 = 3.90$, $J_2 = -7.90$, $J_2' = -1.21$, and $K = -0.73$ meV, exactly favoring the spin configuration illustrated in Fig. 1(a). Since the bond-

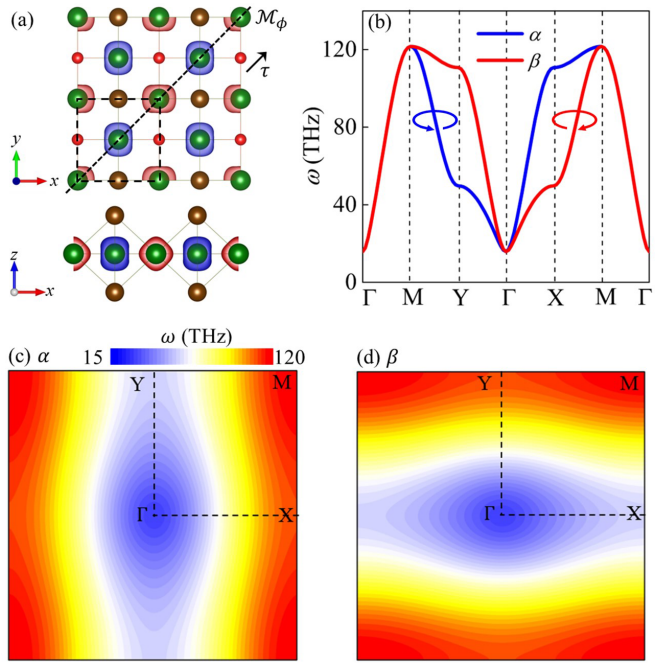


FIG. 2. (a) The top and side views of the crystal structure of CTO. The dashed square indicates a unit cell with the optimized lattice constant $a = 4.12$ Å. The bonding angle of Cr–Te–Cr and Cr-O-Cr equal 94° and 180°, respectively. The blue and red zones located around Cr elucidate the spin density distribution in the real space. (b) Magnon band structure with spin splitting. (c) and (d) Magnon spectrum of α and β modes in the whole Brillouin zone.

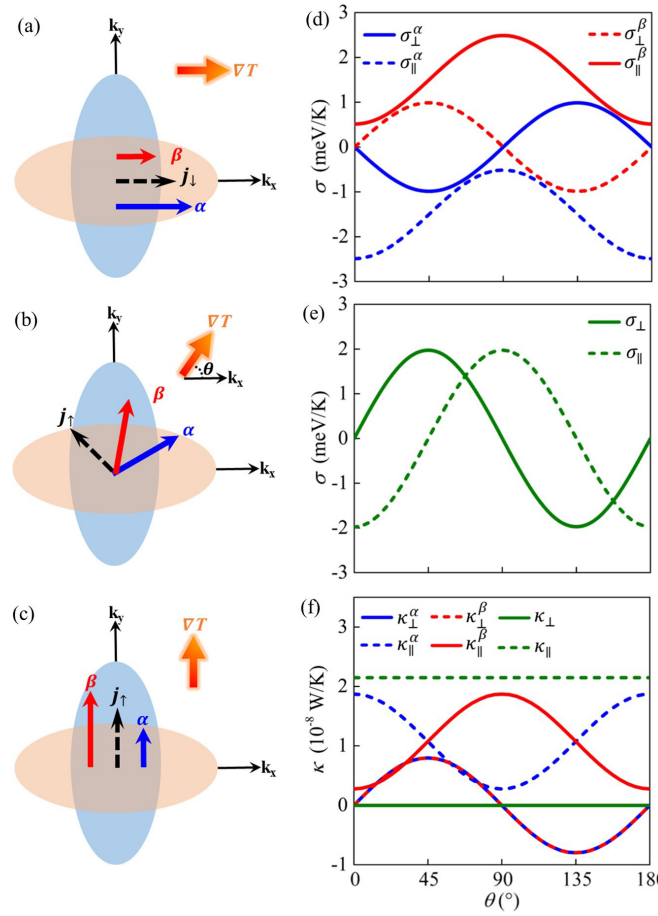


FIG. 3. [(a)–(c)] The flow of spin-polarized magnons driven by different temperature gradients. When the gradient direction is away from the main crystal axis (a) $\hat{\mathbf{x}}$ and (c) $\hat{\mathbf{y}}$, the currents carried by α and β magnons are no longer parallel to its orientation, leading to the transverse spin current in (b). (d) and (e) show the calculated spin conductivity and (f) is thermal conductivity. $\tau_0 = 10$ ps and $T = 100$ K.

ing angle of Cr–Te–Cr is close to 90°, according to GKA rules, the spin superexchange mediated by intervening Te prefers to be ferromagnetic, while the bonding angle of Cr–O–Cr reaching 180° favors the antiferromagnetic superexchange, which is responsible for the difference between J_2 and J_2'. Monte Carlo simulations based on the explicit spin Hamiltonian [Eq. (1)] determine the Néel temperature reaching 205 K [61].

Figure 2(b) plots the magnon spectrum of CTO. Except for $M \leftrightarrow \Gamma$ lines, the frequency of α and β modes splits significantly with the largest splitting of up to 63.39 THz appearing at X and Y points. Magnonic anisotropy is more clearly illustrated in Figs. 2(c) and 2(d). For the α mode, the more rapid frequency variation along $\hat{\mathbf{x}}$ than $\hat{\mathbf{y}}$ implies the much larger group velocity v_x than v_y [61], leading to thermal orientation-dependent magnonic transport behavior, see below. The situation is similar for the β mode but the rapid frequency variation emerges along $\hat{\mathbf{y}}$. Since the two modes carry opposite spin, the above results reveal that the altermagnetic magnon nondegeneracy is characterized by anisotropic spin-momentum locking. Crucially, CTO exhibits insulating features as confirmed by the indirect electronic gap of 0.30 eV [61], indicating negligible thermal electron excitation at low temperatures. Without electronic contribution, the energy dissipation via joule heating can be avoided, and more importantly, the observation and application of thermal/spin current contributed by pristine magnons become possible.

We also find alternate insulating chromium oxyselenide, Cr_2Se_2O monolayer with magnetic parameters $J_1 = 12.48$, $J_2 = -5.80$, $J_2' = 6.69$, and $K = -0.30$ meV holds magnon modes with anisotropic spin-momentum locking [61]. Notably, the DMI is forbidden for proposed candidates based on the Moriya symmetry rules [62]. The phonon dispersion without any imaginary frequency in the entire Brillouin zone and *ab initio* molecular dynamic simulations showing the stable internal energy evolution at 500 K [61] demonstrates their dynamical and thermal stability.

Spin Seebeck and spin nernst effects. The magnon current can be conveniently driven by the temperature or chemical potential bias. In the linear response regime, such spin current density under the temperature T gradient reads

$$\begin{pmatrix} j_x^z \\ j_y^z \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & 0 \\ 0 & \sigma_{yy} \end{pmatrix} \begin{pmatrix} (-\partial_x T)\cos\theta \\ (-\partial_y T)\sin\theta \end{pmatrix}, \tag{6}$$

where θ is the direction of the temperature gradient with respect to the $\hat{\mathbf{x}}$ direction [Fig. 3(b)]. Based on the Kubo formula [39,63–66], the magnon thermal conductivity tensor $\sigma_{mn} = \sigma_{mn}^\alpha + \sigma_{mn}^\beta$ is contributed by the α and β mode

separately with [61]

$$\sigma_{mn}^{\alpha} = -\frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}(\hat{V}_m)_{\alpha\alpha}(\hat{V}_n)_{\alpha\alpha}\omega_{\mathbf{k}}^{\alpha}\frac{e^{\hbar\omega_{\mathbf{k}}^{\alpha}/k_BT}}{(e^{\hbar\omega_{\mathbf{k}}^{\alpha}/k_BT}-1)^2},$$

$$\sigma_{mn}^{\beta} = \frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}(\hat{V}_m)_{\beta\beta}(\hat{V}_n)_{\beta\beta}\omega_{\mathbf{k}}^{\beta}\frac{e^{-\hbar\omega_{\mathbf{k}}^{\beta}/k_BT}}{(e^{-\hbar\omega_{\mathbf{k}}^{\beta}/k_BT}-1)^2},$$

where n is along the temperature gradient, m represents the direction of driven spin current, A is the sample area, τ_0 is magnon lifetime, and $\hat{V}_m = T_{\mathbf{k}}^{\dagger}(\partial_{k_m}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}$. The same results are obtained by the semiclassical Boltzmann transport theory with relaxation-time approximation [61]. Notably, the spin Berry curvature $\Omega(\mathbf{k})$ vanishes due to the coexistence of $\mathcal{P}$ and $\mathcal{T}c_x$ symmetries in our altermagnet model [15], where c_x represents 180° rotation operation around the $\hat{\mathbf{x}}$-axis. Taking $\tau_0 = 10$ ps [67–69] and $T = 100$ K, we find $\sigma_{xx}^{\alpha} = -\sigma_{yy}^{\beta} = -2.49$ and $\sigma_{yy}^{\alpha} = -\sigma_{xx}^{\beta} = -0.51$ meV/K in CTO.

By projecting to the gradient direction, we find the longitudinal and transverse spin currents

$$\begin{pmatrix} j_{\parallel}^{z} \\ j_{\perp}^{z} \end{pmatrix} = \begin{pmatrix} \sigma_{\parallel} & 0 \\ 0 & \sigma_{\perp} \end{pmatrix}\begin{pmatrix} -\partial_n T \\ -\partial_n T \end{pmatrix}, \tag{7}$$

where $\sigma_{\parallel} = \sigma_{\parallel}^{\alpha} + \sigma_{\parallel}^{\beta}$ and $\sigma_{\perp} = \sigma_{\perp}^{\beta} - \sigma_{\perp}^{\alpha}$ with $\sigma_{\parallel}^{\alpha} = \sigma_{xx}^{\alpha}\cos^2\theta + \sigma_{yy}^{\alpha}\sin^2\theta$, $\sigma_{\parallel}^{\beta} = \sigma_{xx}^{\beta}\cos^2\theta + \sigma_{yy}^{\beta}\sin^2\theta$, $\sigma_{\perp}^{\alpha} = (\sigma_{xx}^{\alpha} - \sigma_{yy}^{\alpha})\cos\theta\sin\theta$, and $\sigma_{\perp}^{\beta} = (\sigma_{yy}^{\beta} - \sigma_{xx}^{\beta})\cos\theta\sin\theta$. When the temperature gradient is along the main crystal axis $\hat{\mathbf{x}}$ ($\theta = 0°$), only the longitudinal spin current is generated [Figs. 3(a) and 3(d)]. The nondegeneracy of magnon modes leads to $|\sigma_{xx}^{\alpha}| \neq |\sigma_{xx}^{\beta}|$ and thus generates the SSE with a spin-down current $\sigma_{\parallel} = -1.98$ meV/K [Fig. 3(e)]. The external field generating magnon imbalance is not a prerequisite for realizing SSE in altermagnets, however. When the temperature gradient deviates from $\hat{\mathbf{x}}$ ($0° < \theta < 90°$), the currents carried by α and β modes are not parallel to temperature gradient because of the anisotropy $\sigma_{xx}^{\alpha(\beta)} \neq \sigma_{yy}^{\alpha(\beta)}$, and flow in different directions. $|\sigma_{xx(yy)}^{\alpha}| = |\sigma_{yy(xx)}^{\beta}|$ by $\mathcal{M}_{\phi}$ symmetry [Figs. 3(b) and 3(d)]. Consequently, the SNE with transverse spin-up current appears [Fig. 3(e)]. For $\theta = 45°$, the transverse current reaches its maximum with $\sigma_{\perp} = 1.98$ meV/K since the α and β magnons flowing along the opposite directions has maximum magnitudes. The transverse spin current vanishes again as the temperature gradient rotates to the main crystal axis $\hat{\mathbf{y}}$ ($\theta = 90°$), while the longitudinal spin-up current with $\sigma_{\parallel} = 1.98$ meV/K emerges [Figs. 3(c) and 3(e)]. Different from spin current, the magnonic thermal current only flows along the temperature gradient with constant $\kappa_{\perp} = 0$ and $\kappa_{\parallel} = 2.15 \times 10^{-8}$ W/K [Fig. 3(f)], revealing the vanished thermal Hall effect.

Figure 4 addresses $\sigma_{\parallel}$ and $\sigma_{\perp}$ as a function of temperature and $J_2 - J_2'$. Their magnitudes increase rapidly and monotonically with temperature or difference between J_2 and J_2'. This is substantiated since higher temperature produces more magnon occupation by the Bose-Einstein distribution and the larger magnitude of $J_2 - J_2'$ results in strong magnonic anisotropy. Monte Carlo simulations unveil that the normalized magnetization remains around 0.85 [Fig. S2(a)] at the temperature up to 110 K (Fig. 4), indicating the fluctuation around the long-range magnetic order is still small. This

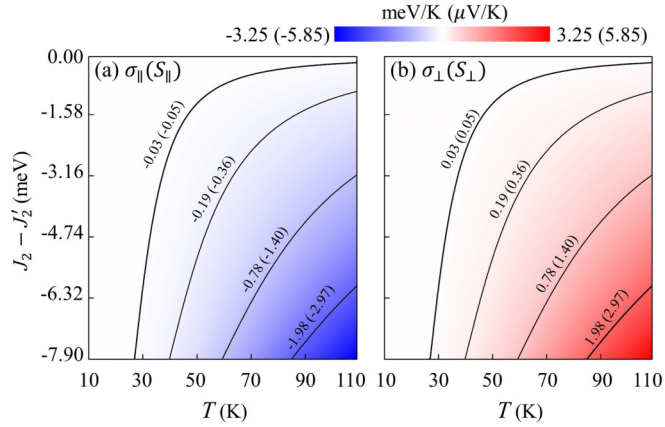


FIG. 4. (a) SSE and (b) SNE coefficients as a function of temperature and $J_2 - J_2'$. The temperature gradient is set along the $\hat{\mathbf{x}}$ and diagonal directions for observing SSE and SNE, respectively.

validates the model of altermagnetic magnons based on the linear spin wave theory when the temperature range is much lower than the critical one. The thermal fluctuation will break the long-range magnetic order when temperature increases to 300 K, thus leading to the vanishment of SSE and SNE. Even at low temperature $T = 29$ K, the SNE $\sigma_{\perp}$ reaches 0.03 meV/K that is two orders in magnitude larger than that induced by nonzero spin Berry curvature in monolayer and bilayer antiferromagnets [15,33,34]. Further, we compare the SSE coefficients in experiments by adopting the formula $\mathcal{S} = (e/\hbar d)\rho\theta_{\mathrm{SH}}\sigma\cos(\eta)$ that provides the inverse spin Hall voltage [35,70,71], where d is the interlayer distance, the electrical resistance $\rho = 105$ nΩ m, the spin Hall angle of Pt $\theta_{\mathrm{SH}} = 0.068$ [72], and the bulk inclination angle $\eta = 45°$. We find that SSE is of the same order as the experimental record of antiferromagnets, where a magnetic field around 10 T is required [23,25,73,74].

Conclusion and discussion. We predict the SSE and SNE in collinearly antiferromagnetic insulators with efficient heat-to-spin conversion, overcoming the limitations from Joule heat, external magnetic field, and SOC. The spin Seebeck conductivity is of the same magnitude as the records in other antiferromagnets where a large magnetic field is desired for generating spin splitting, and the spin Nernst conductivity is much larger than that of antiferromagnetic few layers induced by spin Berry curvature. Within an appropriate temperature range, both spin Seebeck and Nernst conductivities can be significantly enhanced by increasing temperature. The strong longitudinal and transverse spin currents driven by thermal bias can be further applied to produce torque on adjacent magnets, thus providing a way for manipulating magnetization. The anisotropic spin-momentum locking of magnons, arising from the anisotropic exchange coupling determined by $\mathcal{P}\mathcal{T}\tau$ symmetry breaking and $\mathcal{M}_{\phi}\tau$ symmetry preservation, results in a distinct response of α and β magnons. Meanwhile, the exchange coupling is normally larger than the SOC-induced DMI since SOC is a small perturbation. The consequent spin currents thus obviously exceed that of the collinear magnets generated by the DMI. The coexistence of $\mathcal{P}$ and $\mathcal{T}c_x$ symmetries excludes any role of Berry curvature on magnon propagation. Possible material candidates are

also provided for realizing proposed phenomena that should widely exist in altermagnetic insulators [61] with easy-axis anisotropy.

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[1] Y. Kajiwara, K. Harii, S. Takahashi, J. Ohe, K. Uchida, M. Mizuguchi, H. Umezawa, H. Kawai, K. Ando, K. Takanashi, S. Maekawa, and E. Saitoh, Transmission of electrical signals by spin-wave interconversion in a magnetic insulator, Nature (London) **464**, 262 (2010).

[2] G. Bauer, E. Saitoh, and B. van Wees, Spin caloritronics, Nat. Mater. **11**, 391 (2012).

[3] A. V. Chumak, A. A. Serga, and B. Hillebrands, Magnon transistor for all-magnon data processing, Nat. Commun. **5**, 4700 (2014).

[4] A. V. Chumak, V. I. Vasyuchka, A. A. Serga, and B. Hillebrands, Magnon spintronics, Nat. Phys. **11**, 453 (2015).

[5] L. J. Cornelissen, J. Liu, R. A. Duine, J. Ben Youssef, and B. J. van Wees, Long-distance transport of magnon spin information in a magnetic insulator at room temperature, Nat. Phys. **11**, 1022 (2015).

[6] T. Yu, Z. Luo, and G. Bauer, Chirality as generalized spin–orbit interaction in spintronics, Phys. Rep. **1009**, 1 (2023).

[7] W. Xing, L. Qiu, X. Wang, Y. Yao, Y. Ma, R. Cai, S. Jia, X. C. Xie, and W. Han, Magnon transport in quasi-two-dimensional van der waals antiferromagnets, Phys. Rev. X **9**, 011026 (2019).

[8] Q. Wang, M. Kewenig, M. Schneider, R. Verba, F. Kohl, B. Heinz, M. Geilen, M. Mohseni, B. Lägel, F. Ciubotaru, C. Adelmann, C. Dubs, S. D. Cotofana, O. V. Dobrovolskiy, T. Brächer, P. Pirro, and A. V. Chumak, A magnonic directional coupler for integrated magnonic half-adders, Nat. Electron. **3**, 765 (2020).

[9] G. Chen, S. Qi, J. Liu, D. Chen, J. Wang, S. Yan, Y. Zhang, S. Cao, M. Lu, S. Tian, K. Chen, P. Yu, Z. Liu, X. C. Xie, J. Xiao, R. Shindou, and J. Chen, Electrically switchable van der Waals magnon valves, Nat. Commun. **12**, 6279 (2021).

[10] X.-Y. Wei, O. A. Santos, C. H. S. Lusero, G. E. W. Bauer, J. B. Youssef, and B. J. van Wees, Giant magnon spin conductivity in ultrathin yttrium iron garnet films, Nat. Mater. **21**, 1352 (2022).

[11] J. Li, C. B. Wilson, R. Cheng, M. Lohmann, M. Kavand, W. Yuan, M. Aldosary, N. Agladze, P. Wei, M. S. Sherwin, and Jing Shi, Spin current from sub-terahertz-generated antiferromagnetic magnons, Nature (London) **578**, 70 (2020).

[12] S. M. Rezende, R. L. Rodríguez-Suárez, and A. Azevedo, Diffusive magnonic spin transport in antiferromagnetic insulators, Phys. Rev. B **93**, 054412 (2016).

[13] S. M. Rezende, R. L. Rodríguez-Suárez, and A. Azevedo, Theory of spin Seebeck effects in antiferromagnets, Phys. Rev. B **93**, 014425 (2016).

[14] K. Chen, W. Lin, C. L. Chien, and S. Zhang, Temperature dependence of angular momentum transport across interfaces, Phys. Rev. B **94**, 054413 (2016).

[15] R. Cheng, S. Okamoto, and D. Xiao, Spin nernst effect of magnons in collinear antiferromagnets, Phys. Rev. Lett. **117**, 217202 (2016).

[16] R. Cheng, D. Xiao, and J.-G. Zhu, Interlayer couplings mediated by antiferromagnetic magnons, Phys. Rev. Lett. **121**, 207202 (2018).

[17] K. Shen, Magnon spin relaxation and spin hall effect due to the dipolar interaction in antiferromagnetic insulators, Phys. Rev. Lett. **124**, 077201 (2020).

[18] K. Uchida, S. Takahashi, K. Harii, J. Ieda, W. Koshibae, K. Ando, S. Maekawa, and E. Saitoh, Observation of the spin Seebeck effect, Nature (London) **455**, 778 (2008).

[19] C. M. Jaworski, R. C. Myers, E. Johnston-Halperin, and J. P. Heremans, Giant spin Seebeck effect in a non-magnetic material, Nature (London) **487**, 210 (2012).

[20] K. Uchida, J. Xiao, H. Adachi, J. Ohe, S. Takahashi, J. Ieda, T. Ota, Y. Kajiwara, H. Umezawa, H. Kawai, G. E. W. Bauer, S. Maekawa, and E. Saitoh, Spin seebeck insulator, Nat. Mater. **9**, 894 (2010).

[21] S. Meyer, Y.-T. Chen, S. Wimmer, M. Althammer, T. Wimmer, R. Schlitz, S. Geprägs, H. Huebl, D. Ködderitzsch, H. Ebert, G. E. W. Bauer, R. Gross, and S. T. B. Goennenwein, Observation of the spin Nernst effect, Nat. Mater. **16**, 977 (2017).

[22] P. Sheng, Y. Sakuraba, Y. Lau, S. Takahashi, S. Mitani, and M. Hayashi, The spin nernst effect in tungsten, Sci. Adv. **3**, e1701503 (2017).

[23] S. M. Wu, W. Zhang, A. KC, P. Borisov, J. E. Pearson, J. S. Jiang, D. Lederman, A. Hoffmann, and A. Bhattacharya, Antiferromagnetic spin seebeck effect, Phys. Rev. Lett. **116**, 097204 (2016).

[24] W. Lin, K. Chen, S. Zhang, and C. L. Chien, Enhancement of thermally injected spin current through an Antiferromagnetic insulator, Phys. Rev. Lett. **116**, 186601 (2016).

[25] S. Seki, T. Ideue, M. Kubota, Y. Kozuka, R. Takagi, M. Nakamura, Y. Kaneko, M. Kawasaki, and Y. Tokura, Thermal generation of spin current in an antiferromagnet, Phys. Rev. Lett. **115**, 266601 (2015).

[26] J. Li, Z. Shi, V. H. Ortiz, M. Aldosary, C. Chen, V. Aji, P. Wei, and J. Shi, Spin seebeck effect from antiferromagnetic magnons and critical spin fluctuations in epitaxial FeF_2 films, Phys. Rev. Lett. **122**, 217204 (2019).

[27] V. A. Zyuzin and A. A. Kovalev, Magnon spin nernst effect in antiferromagnets, Phys. Rev. Lett. **117**, 217203 (2016).

[28] T. Kikkawa, K. Shen, B. Flebus, R. A. Duine, K. Uchida, Z. Qiu, G. E. W. Bauer, and E. Saitoh, Magnon polarons in the spin seebeck effect, Phys. Rev. Lett. **117**, 207203 (2016).

[29] D. Qu, S. Y. Huang, J. Hu, R. Wu, and C. L. Chien, Intrinsic spin seebeck effect in Au/YIG, Phys. Rev. Lett. **110**, 067206 (2013).

[30] J. Xu, J. He, J.-S. Zhou, D. Qu, S. Huang, and C. L. Chien, Observation of vector spin seebeck effect in a noncollinear antiferromagnet, Phys. Rev. Lett. **129**, 117202 (2022).

[31] K. Tauber, M. Gradhand, D. V. Fedorov, and I. Mertig, Extrinsic spin nernst effect from first principles, Phys. Rev. Lett. **109**, 026601 (2012).

[32] J. Xiao, G. E. W. Bauer, K. Uchida, E. Saitoh, and S. Maekawa, Theory of magnon-driven spin Seebeck effect, Phys. Rev. B **81**, 214418 (2010).

[33] S. Park, N. Nagosa, and B.-J. Yang, thermal Hall effect, spin Nernst effect, and Spin density induced by a thermal gradient in collinear ferrimagnets from magnon-phonon interaction, Nano Lett. **20**, 2741 (2020).

[34] G. Go and S. K. Kim, Tunable large spin Nernst effect in a two-dimensional magnetic bilayer, Phys. Rev. B **106**, 125103 (2022).

[35] Y. Shiomi, R. Takashima, and E. Saitoh, Experimental evidence consistent with a magnon Nernst effect in the Antiferromagnetic insulator $MnPS_3$, Phys. Rev. B **96**, 134425 (2017).

[36] P. R. T. Ribeiro, F. L. A. Machado, M. Gamino, A. Azevedo, and S. M. Rezende, Spin Seebeck effect in antiferromagnet nickel oxide in wide ranges of temperature and magnetic field, Phys. Rev. B **99**, 094432 (2019).

[37] Q. Cui, L. Wang, Y. Zhu, J. Liang, and H. Yang, Magnetic anisotropy, exchange coupling and Dzyaloshinskii-Moriya interaction of two-dimensional magnets, Front. Phys. **18**, 13602 (2023).

[38] H. Yang, J. Liang, and Q. Cui, First-principles calculations for Dzyaloshinskii–Moriya interaction, Nat. Rev. Phys. **5**, 43 (2023).

[39] A. Mook, R. R. Neumann, J. Henk, and I. Mertig, Spin Seebeck and spin Nernst effects of magnons in noncollinear Antiferromagnetic insulators, Phys. Rev. B **100**, 100401(R) (2019).

[40] B. Li, S. Sandhoefner, and A. A. Kovalev, Intrinsic spin Nernst effect of magnons in a noncollinear antiferromagnet, Phys. Rev. Res. **2**, 013079 (2020).

[41] S. Hayami, Y. Yanagi, and H. Kusunose, Momentum-dependent spin splitting by collinear antiferromagnetic ordering, J. Phys. Soc. Jpn. **88**, 123702 (2019).

[42] S. Hayami, Y. Yanagi, and H. Kusunose, Bottom-up design of spin-split and reshaped electronic band structures in antiferromagnets without spin-orbit coupling: Procedure on the basis of augmented multipoles, Phys. Rev. B **102**, 144441 (2020).

[43] L. Šmejkal, R. González-Hernández, T. Jungwirth, and J. Sinova, Crystal time-reversal symmetry breaking and spontaneous Hall effect in collinear antiferromagnets, Sci. Adv. **6**, eaaz8809 (2020).

[44] L. Šmejkal, J. Sinova, and T. Jungwirth, Beyond conventional ferromagnetism and antiferromagnetism: A phase with nonrelativistic spin and crystal rotation symmetry, Phys. Rev. X **12**, 031042 (2022).

[45] L. Šmejkal, J. Sinova, and T. Jungwirth, Emerging research landscape of altermagnetism, Phys. Rev. X **12**, 040501 (2022).

[46] L. Yuan, Z. Wang, J. Luo, and A. Zunger, Strong influence of nonmagnetic ligands on the momentum-dependent spin splitting in antiferromagnets, Phys. Rev. B **103**, 224410 (2021).

[47] R. He, D. Wang, N. Luo, J. Zeng, K.-Q. Chen, and L.-M. Tang, Nonrelativistic spin-momentum coupling in antiferromagnetic twisted bilayers, Phys. Rev. Lett. **130**, 046401 (2023).

[48] H. Bai, L. Han, X. Y. Feng, Y. J. Zhou, R. X. Su, Q. Wang, L. Y. Liao, W. X. Zhu, X. Z. Chen, F. Pan, X. L. Fan, and C. Song, Observation of spin splitting torque in a collinear antiferromagnet RuO_2, Phys. Rev. Lett. **128**, 197202 (2022).

[49] S. Karube, T. Tanaka, D. Sugawara, N. Kadoguchi, M. Kohda, and J. Nitta, Observation of spin-splitter torque in collinear antiferromagnetic RuO_2, Phys. Rev. Lett. **129**, 137201 (2022).

[50] A. Bose, N. J. Schreiber, R. Jain, D.-F. Shao, H. P. Nair, J. Sun, X. S. Zhang, D. A. Muller, E. Y. Tsymbal, D. G. Schlom, and D. C. Ralph, Tilted spin current generated by an antiferromagnet, Nat. Electron. **5**, 263 (2022).

[51] L. Šmejkal, A. Marmodoro, K.-H. Ahn, R. G.-Hernández, I. Turek, S. Mankovsky, H. Ebert, S. W. D'Souza, O. Šipr, J. Sinova, and T. Jungwirth, Chiral magnons in altermagnetic RuO2, arXiv:2211.13806 [Phys. Rev. Lett. (to be published)].

[52] V. Baltz, A. Manchon, M. Tsoi, T. Moriyama, T. Ono, and Y. Tserkovnyak, Antiferromagnetic spintronics, Rev. Mod. Phys. **90**, 015005 (2018).

[53] H.-Y. Ma, M. Hu, N. Li, J. Liu, W. Yao, J.-F. Jia, and J. Liu, Multifunctional Antiferromagnetic materials with giant piezomagnetism and noncollinear spin current, Nat. Commun. **12**, 2846 (2021).

[54] B. Goodenough, Theory of the role of covalence in the perovskite-type manganites [La, M(II)]MnO_3, Phys. Rev. **100**, 564 (1955).

[55] J. Kanamori, Superexchange interaction and symmetry properties of electron orbitals, J. Phys. Chem. Solids **10**, 87 (1959).

[56] P. W. Anderson, New approach to the theory of superexchange interactions, Phys. Rev. **115**, 2 (1959).

[57] T. Holstein and H. Primakoff, Field dependence of the intrinsic domain magnetization of a ferromagnet, Phys. Rev. **58**, 1098 (1940).

[58] N. D. Mermin and H. Wagner, Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic heisenberg models, Phys. Rev. Lett. **17**, 1133 (1966).

[59] C. Gong, L. Li, Z. Li, H. Ji, A. Stern, Y. Xia, T. Cao, W. Bao, C. Wang, Y. Wang, Z. Q. Qiu, R. J. Cava, S. G. Louie, J. Xia, and X. Zhang, Discovery of intrinsic ferromagnetism in two-dimensional van der Waals crystals, Nature (London) **546**, 265 (2017).

[60] B. Huang, G. Clark, E. Navarro-Moratalla, D. R. Klein, R. Cheng, K. L. Seyler, D. Zhong, E. Schmidgall, M. A. McGuire, D. H. Cobden, W. Yao, D. Xiao, P. J. Herrero, and X. Xu, Layer-dependent ferromagnetism in a van der Waals crystal down to the monolayer limit, Nature (London) **546**, 270 (2017).

[61] See Supplemental Material at http://link.aps.org/supplemental/10.1103/PhysRevB.108.L180401 for (i) construction of the magnons Hamiltonian, (ii) details of first-principles calculations, (iii) Kubo formula for spin/thermal conductivity, (iv) Boltzmann transport theory for spin/thermal conductivity, (v) magnetic and structural properties of proposed candidates, (vi) group velocity of altermagnetic magnons, (vii) magnon dispersion of CSO, and (viii) other potential altermagnets for realizing spin thermal transport, which also includes Refs. [27,33,39,57,63–65,75–82].

[62] T. Moriya, Anisotropic superexchange interaction and weak ferromagnetism, Phys. Rev. **120**, 91 (1960).

[63] R. Kubo, M. Yokota, and S. Nakajima, Statistical-mechanical theory of irreversible processes. ii. response to thermal disturbance, J. Phys. Soc. Jpn. **12**, 1203 (1957).

[64] G. D. Mahan, *Many-Particle Physics* (Springer US, Boston, MA, 2000).

[65] R. Matsumoto, R. Shindou, and S. Murakami, Thermal Hall effect of magnons in magnets with dipolar interaction, Phys. Rev. B **89**, 054420 (2014).

[66] T. Yu, S. Sharma, Y. M. Blanter, and G. E. W. Bauer, Surface dynamics of rough magnetic films, Phys. Rev. B **99**, 174402 (2019).

[67] Magnon lifetime is estimated to be on the order of 10 ps based on the Raman spectra of two-dimensional collinear antiferromagnets.

[68] A. McCreary, J. R. Simpson, T. T. Mai, R. D. McMichael, J. E. Douglas, N. Butch, C. Dennis, R. V. Aguilar, and A. R. H. Walker, Quasi-two-dimensional magnon identification in antiferromagnetic $FePS_3$ via magneto-Raman spectroscopy, Phys. Rev. B **101**, 064416 (2020).

[69] T. T. Mai, K. F. Garrity, A. McCreary, J. Argo, J. R. Simpson, V. Doan-Nguyen, R. V. Aguilar, and A. R. H. Walker, Magnon-phonon hybridization in two-dimensional antiferromagnet $MnPSe_3$, Sci. Adv. **7**, eabj3106 (2021).

[70] N. Bazazzadeh, M. Hamdi, F. Haddadi, A. Khavasi, A. Sadeghi, and S. M. Mohseni, Symmetry enhanced spin-Nernst effect in honeycomb Antiferromagnetic transition metal trichalcogenide monolayers, Phys. Rev. B **103**, 014425 (2021).

[71] N. Bazazzadeh, M. Hamdi, S. Park, A. Khavasi, S. M. Mohseni, and A. Sadeghi, Magnetoelastic coupling enabled tunability of magnon spin current generation in two-dimensional antiferromagnets, Phys. Rev. B **104**, L180402 (2021).

[72] Y. Wang, P. Deorani, X. Qiu, J. H. Kwon, and H. Yang, Determination of intrinsic spin Hall angle in Pt, Appl. Phys. Lett. **105**, 152412 (2014).

[73] T. Kikkawa, K. Uchida, S. Daimon, Z. Qiu, Y. Shiomi, and E. Saitoh, Critical suppression of spin Seebeck effect by magnetic fields, Phys. Rev. B **92**, 064413 (2015).

[74] Y. Shiomi, R. Takashima, D. Okuyama, G. Gitgeatpong, P. Piyawongwatthana, K. Matan, T. J. Sato, and E. Saitoh, Spin Seebeck effect in the polar antiferromagnet $\alpha - Cu_2V_2O_7$, Phys. Rev. B **96**, 180414(R) (2017).

[75] G. Kresse and J. Hafner, Ab initio molecular-dynamics simulation of the liquid-metal–amorphous-semiconductor transition in germanium, Phys. Rev. B **49**, 14251 (1994).

[76] G. Kresse and J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys. Rev. B **54**, 11169 (1996).

[77] G. Kresse and D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, Phys. Rev. B **59**, 1758 (1999).

[78] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, Phys. Rev. Lett. **77**, 3865 (1996).

[79] A. Togo and I. Tanaka, First principles phonon calculations in materials science, Scr. Mater. **108**, 1 (2015).

[80] S. Nosé, A unified formulation of the constant temperature molecular dynamics methods, J. Chem. Phys. **81**, 511 (1984).

[81] Q. Cui, J. Liang, Y. Zhu, X. Yao, and H. Yang, Quantum anomalous Hall effects controlled by chiral domain walls, Chinese Phys. Lett. **40**, 037502 (2023).

[82] Q. Cui, Y. Zhu, Y. Ga, J. Liang, P. Li, D. Yu, P. Cui, and H. Yang, Anisotropic dzyaloshinskii–moriya interaction and topological magnetism in two-dimensional magnets protected by P-4m2 crystal symmetry, Nano Lett. **22**, 2334 (2022).